Week 06: Programming and Research

Programming task

Session 2: deepen this week’s Python.
Predict → change → repair.
Session 3: apply it to the project.

Research task

In class: discuss Research 4.
After class: Research 5 on failed updates.
Prepare your evidence; discuss in Week 7.

Open learning_materials/Week_06.ipynb. Follow Start Here; keep your work in that notebook. Email: submit.progmis@gmail.com.

Programming task

Session 2 · Python practice before project application.

1. Predict two objects

Do this

Open #session-2: Predict → Run.
Read Counter before executing.
Predict both doubled() results.

Observe and explain

Identify the class and two instances.
Name count and doubled().
Explain which instance is self.

2. Change one attribute

Do this

Copy the example.
Set first.count = 5.
Predict both returned values; run.

Observe and explain

Expected: first returns 10; second 4.
Explain why only one object changes.
Keep prediction and observed output.

3. Repair the method

Do this

Run BuggyCounter with count 4.
It adds 2 instead of doubling.
Repair doubled() in your own copy.

Observe and explain

Check counts 4 and 1.
Explain why count 2 hides the bug.
Optional: compare alias = first with second.

Programming practice: done checklist

Keep in your notebook

Prediction before running
Your changed or repaired code
Observed output or error
One explanation of why it happened

Next step

Optional exploration can wait.
Save the notebook and a backup.
Continue to the project application.
Submit to submit.progmis@gmail.com.

Outside class: complete the practice and Research preparation.

Project application: save and recover P11

Step	What to do / check
Open Session 3	#project-task: use the supplied SQLite setup; record the database path.
Save and reopen	P11, planning: Q11×2 and Q12×1; save unit snapshots; commit, close, reconnect.
Compare results	Old estimate 11; change Q11 unit 3→4: current estimate 13, saved estimate still 11.
Check and keep	Unknown ID handled; stock stays 10/4. Keep rows, relationship sketch, peer note and explanation.

Save the notebook outputs before cleaning up its temporary database.

Research database transactions

What to research

Find out what a transaction is and why related updates should succeed or fail together.

Possible directions

Use an action with two updates. Consider a failure between them and how to check the records before and after.

A short summary example

A reservation reduces stock and saves a request. If saving fails, undo the stock change. Compare both records before and after.

Discuss in Week 7 · summarize in your own words and cite your sources.